leetcode.com/problems/distinct-subsequences/

Problem List | Submit | 00:00:00 | Premium

Description | Editorial | Solutions | Submissions

115. Distinct Subsequences

Hard | Topics | Companies

Given two strings s and t, return *the number of distinct* ***subsequences*** *of s which equals* t.

The test cases are generated so that the answer fits on a 32-bit signed integer.

Example 1:

```
Input: s = "rabbbit", t = "rabbit"
Output: 3
Explanation:
As shown below, there are 3 ways you can generate "rabbit" from s.
rabbbit
rabbbit
rabbbit
```

Example 2:

```
Input: s = "babgbag", t = "bag"
Output: 5
Explanation:
As shown below, there are 5 ways you can generate "bag" from s.
```

7.4K | 244 | 48 Online

Code

C++ | Auto | Format

```cpp
class Solution {
public:
    int numDistinct(string s, string t) {
        int rows = s.size(), cols = t.size();
        vector<vector<long long>> table(rows + 1, vector<long long>(cols + 1, 0));

        for (int i = 0; i <= rows; i++)
            table[i][0] = 1;

        for (int i = 1; i <= rows; i++) {
            for (int j = 1; j <= cols; j++) {
                table[i][j] = table[i-1][j];
                if (s[i-1] == t[j-1])
                    table[i][j] += table[i-1][j-1];
            }
        }

        return table[rows][cols];
    }
};
```

Saved | Ln 20, Col 3

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2

Input

s =
"rabbbit"

t =
"rabbit"

Output

3

Expected

3

Contribute a testcase

leetcode.com/problems/distinct-subsequences/submissions/1986927665/

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115. Distinct Subsequences

Solved

Hard | Topics | Companies

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Example 2:

```
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Output: 5
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Accepted

All Submissions

Accepted 66 / 66 testcases passed

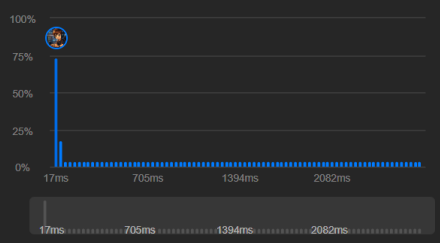
Arnav_70131 submitted at Apr 24, 2026 16:31

Runtime

31 ms | Beats 52.58%

Memory

44.52 MB | Beats 12.19%



Code | C++

```cpp
class Solution {
public:
```

Code

C++ | Auto | Format

```cpp
class Solution {
public:
    int numDistinct(string s, string t) {
        int rows = s.size(), cols = t.size();
        vector<vector<double>> table(rows + 1, vector<double>(cols + 1, 0));

        for (int i = 0; i <= rows; i++)
            table[i][0] = 1;

        for (int i = 1; i <= rows; i++) {
            for (int j = 1; j <= cols; j++) {
                table[i][j] = table[i-1][j];
                if (s[i-1] == t[j-1])
                    table[i][j] += table[i-1][j-1];
            }
        }

        return (int)table[rows][cols];
    }
};
```

Saved | Ln 20, Col 3

Test Resu

Accepted

Case 1

Input

s =
"rabbbit"

t =
"rabbit"

Output

3

Expected

3